Computational Finance — Lecture 3

Implied Volatility

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Lecture 3 - Implied Volatility

- Recap of Key Results
- Black-Scholes Implied Volatility
- Bisection Method
- Newton-Raphson Method
- Implied Volatility Surface
- Deficiencies of the Black-Scholes Model

Key results: BS Model

We have learned that in the Black-Scholes model with two tradable assets: the bank account M and a risky security (e.g., stock) S,

$$dM_t = rM_t dt$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

the price of a contingent clain that is paid at time T in the form of $V(S_T, T) = g(S_T)$, is a function $V(S_t, t)$ satisfying the Black-Scholes PDE

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial^2 S} - rV_t = 0$$

with the terminal contidion

$$V(S_T, T) = g(S_T).$$

Key results: Martingale approach

The price of a contingent claim is given by

$$V(S_t, t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\left[V(S_T, T) \mid S_t\right],$$

where $\mathbb{E}^{\mathbb{Q}}$ denotes the expectation under the risk-neutral measure $\mathbb{Q}$, under which the underlying asset price process satisfies

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t \mathrm{d}W_t^{\mathbb{Q}}.$$

The relative price of the contingent claim with respect to the bank account numeraire M_t, namely

$$\frac{V(S_t, t)}{M_t},$$

is a martingale under the risk-neutral measure $\mathbb{Q}$.

Key results: Feynman-Kac formula

The Feynman–Kac theorem tells that the solution to the PDE

$$\frac{\partial V(x,t)}{\partial t} + \mu(x,t)\frac{\partial V(x,t)}{\partial x} + \frac{1}{2}\sigma^2(x,t)\frac{\partial^2 V(x,t)}{\partial x^2} - r(x,t)V(x,t) + f(x,t) = 0,$$

for $x \in \mathbb{R}$ and $t \in [0, T]$, with terminal condition

$$V(x,T) = g(x),$$

where μ, σ, r, f and g are given functions of x and t, can be written as the conditional expectation

$$V(x,t) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(X_s,s)\,ds} g(X_T) + \int_t^T e^{-\int_t^\tau r(X_s,s)\,ds} f(X_\tau,\tau)\,d\tau \,\middle|\, X_t = x \right],$$

under the probability measure $\mathbb{Q}$ such that the process X_s satisfies

$$dX_s = \mu(X_s,s)\,ds + \sigma(X_s,s)\,dW_s^{\mathbb{Q}}, \qquad s > t,$$

with initial condition $X_t = x$.

Key results: Computational Methods

We can already make a list of possible computational methods for pricing:

- Solving the PDE directly, for example via the finite difference method.
- Computing the risk–neutral expectation

$$V(S_t, t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[V(S_T, T) \mid S_t] = e^{-r(T-t)} \int_0^{+\infty} V(S_T, T)\, f(S_T \mid S_t)\, dS_T,$$

  where $f(S_T \mid S_t)$ denotes the conditional probability density function of S_T given S_t. In many pricing models, the characteristic function corresponding to $f(S_T \mid S_t)$ is available in analytic or semi-analytic form.
- Monte Carlo simulation.
- Numerical integration.
- Fourier-based methods, such as the COS method.

Key results: Analytical Solutions

Simple contingent payoffs admit closed-form solutions in the Black–Scholes model. The price of a European call option with strike K and maturity T is given by the Black–Scholes formula

$$C(S_t, t) = S_t\, N(d_1) - Ke^{-r(T-t)} N(d_2),$$

with

$$d_1 = \frac{\ln(S_t/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = \frac{\ln(S_t/K) + (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}.$$

Implied Volatility of the Black-Scholes Model

There are five parameters in the Black–Scholes model, four of which are known:

- time to maturity $T - t$,
- strike K,
- interest rate r,
- underlying price at time t, S_t.

The unknown parameter is the volatility σ.
We can extract the volatility from observed market data: given a quoted option price and the values of S_t, K, r, and $T - t$, we determine the value of σ that reproduces the market quote.
A volatility computed in this way is called the implied volatility, that is, it is defined as the solution to

$$V^{\mathrm{BS}}(\sigma^{\mathrm{impl}}, r, T - t, K, S_t) - V^{\mathrm{mkt}} = 0.$$

We focus here on extracting σ from a European call option quote. For this purpose, we write the call price as a function of σ, denoted by $C^{\mathrm{BS}}(\sigma)$. An analogous treatment applies to a European put option.

Option quatations in the market

S&P 500 INDEX (^SPX)

Chicago Options - Chicago Options Delayed Price. Currency in USD

Follow

3,970.15 -12.09 (-0.30%)

At close: February 28 05:10PM EST

Summary | Chart | Historical Data | **Options** | Components

April 28, 2023 | In The Money | Show: **List** | Straddle | Option Lookup

Calls for April 28, 2023

Contract Name	Last Trade Date	Strike	Last Price	Bid	Ask	Change	% Change	Volume	Open Interest	Implied Volatility
SPXW230428C02450000	2023-02-22 12:59PM EST	2,450.00	1,571.65	1,537.10	1,542.10	0.00	-	-	0	73.34%
SPXW230428C02900000	2022-12-07 10:59AM EST	2,900.00	1,104.22	996.20	1,007.10	0.00	-	-	1	0.00%
SPXW230428C03050000	2022-12-16 11:14AM EST	3,050.00	860.43	976.00	991.40	0.00	-	2	1	61.74%
SPXW230428C03250000	2022-11-18 3:53PM EST	3,250.00	809.28	385.60	1,155.70	0.00	-	2	1	129.78%
SPXW230428C03325000	2023-01-18 9:42AM EST	3,325.00	726.86	766.00	773.10	0.00	-	2	2	61.55%
SPXW230428C03350000	2022-12-19 1:58AM EST	3,350.00	602.70	695.50	701.30	0.00	-	-	0	50.00%
SPXW230428C03400000	2022-12-22 2:48PM EST	3,400.00	496.20	611.70	637.30	0.00	-	-	1	43.78%
SPXW230428C03425000	2023-01-19 1:03PM EST	3,425.00	544.50	683.10	690.20	0.00	-	2	4	58.96%
SPXW230428C03480000	2023-01-18 10:30AM EST	3,480.00	586.57	613.00	618.70	0.00	-	-	1	52.27%
SPXW230428C03500000	2023-01-23 1:14PM EST	3,500.00	586.57	532.20	538.60	0.00	-	6	5	38.71%
SPXW230428C03525000	2022-11-03 9:04AM EST	3,525.00	399.09	641.40	647.50	0.00	-	-	0	63.67%
SPXW230428C03550000	2023-02-02 3:47PM EST	3,550.00	660.10	463.00	467.50	0.00	-	4	0	31.04%

In the money S&P500 European calls maturing on 2023-Apr-28. Source: Yahoo Finance as of 2023-Mar-01.https://finance.yahoo.com/quote/%5ESPX/options?

Uniqueness of the implied volatility

The Vega of an option is defined as the sensitivity of the option price with respect to the volatility,

$$\mathrm{Vega} := \frac{\partial C}{\partial \sigma}.$$

In the Black–Scholes model, the call price is

$$C(\sigma) = S_t N(d_1) - K e^{-r\tau} N(d_2), \qquad \tau := T - t.$$

We aim to show that $\mathrm{Vega} > 0$, which implies that the call price $C(\sigma)$ is a strictly increasing function of σ.

Uniqueness of the implied volatility

Using the chain rule and $N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, we obtain

$$\frac{\partial C}{\partial \sigma} = S_t N'(d_1) \frac{\partial d_1}{\partial \sigma} - K e^{-r\tau} N'(d_2) \frac{\partial d_2}{\partial \sigma}.$$

Since $d_2 = d_1 - \sigma\sqrt{\tau}$, we have

$$\frac{\partial d_2}{\partial \sigma} = \frac{\partial d_1}{\partial \sigma} - \sqrt{\tau}.$$

The following identity holds in the Black–Scholes model:

$$S_t N'(d_1) = K e^{-r\tau} N'(d_2).$$

It follows from the relation $d_2 = d_1 - \sigma\sqrt{\tau}$ and the exponential form of the normal density $N'(x)$.
Using this identity, all terms involving $\partial d_1 / \partial \sigma$ cancel out.

Uniqueness of the implied volatility

Next, we determine the range of the market option price C^{mkt} for which a solution σ to the equation $C(\sigma) = C^{\mathrm{mkt}}$ exists.

The minimum value of $C(\sigma)$ is given by the limit as $\sigma \to 0^+$.

- If $\ln(S_t/K) + r(T-t) > 0$, then as $\sigma \to 0^+$ we have $d_1 \to +\infty$, $N(d_1) \to 1$, $d_2 \to +\infty$, $N(d_2) \to 1$, hence $C(\sigma) \to S_t - Ke^{-r(T-t)}$.
- If $\ln(S_t/K) + r(T-t) < 0$, then as $\sigma \to 0^+$ we have $d_1 \to -\infty$, $N(d_1) \to 0$, $d_2 \to -\infty$, $N(d_2) \to 0$, hence $C(\sigma) \to 0$.
- If $\ln(S_t/K) + r(T-t) = 0$, then as $\sigma \to 0^+$ we have $d_1 \to 0$, $N(d_1) \to \frac{1}{2}$, $d_2 \to 0$, $N(d_2) \to \frac{1}{2}$, hence $C(\sigma) \to \frac{S_t}{2} - \frac{Ke^{-r(T-t)}}{2} = 0$. The last equality holds because we assumed $\ln(S_t/K) + r(T-t) = 0$.

These three cases can be summarized compactly as

$$\lim_{\sigma \to 0^+} C(\sigma) = \max\big(S_t - Ke^{-r(T-t)}, 0\big).$$

Uniqueness of the implied volatility

The maximum value of $C(\sigma)$ is S_t, i.e.,

$$\lim_{\sigma\to\infty} C(\sigma) = S_t.$$

Indeed, as $\sigma \to \infty$ we have $d_1 \to +\infty$ and $d_2 \to -\infty$, hence

$$\lim_{\sigma\to\infty} N(d_1) = 1, \qquad \lim_{\sigma\to\infty} N(d_2) = 0,$$

and therefore

$$\lim_{\sigma\to\infty} \Big(S_t N(d_1) - Ke^{-r(T-t)} N(d_2) \Big) = S_t.$$

Consequently, for any $\sigma \geq 0$ the call price satisfies the bounds

$$\max\big(S_t - Ke^{-r(T-t)}, 0\big) \leq C(\sigma) < S_t.$$

Therefore, if $C^{\mathrm{mkt}} \in \big[\max(S_t - Ke^{-r(T-t)}, 0), \, S_t\big)$, then there exists a unique implied volatility $\sigma^{\mathrm{impl}} \geq 0$ such that $C(\sigma^{\mathrm{impl}}) = C^{\mathrm{mkt}}$.

Numerical Methods

- Bisection method
- Newton's method

Bisection method

The bisection method is based on the observation that if a continuous function changes sign then it must pass through zero.

For continuous F, if $x_a < x_b$ with $F(x_a)F(x_b) < 0$, then $F(x^*) = 0$ for some $x_a < x^* < x_b$.

Having found x_a and x_b with $F(x_a)F(x_b) < 0$, we could evaluate F at the mid point

$$x_{\mathrm{mid}} = \frac{x_a + x_b}{2}.$$

The sign of $F(x_{\mathrm{mid}})$ must then match either $F(x_a)$ or $F(x_b)$. This means that one of the intervals $[x_a, x_{\mathrm{mid}}]$ or $[x_{\mathrm{mid}}, x_b]$ must contain an x^*.

By repeating this process, we can construct an arbitrarily small interval in which an x^* must lie – hence we can find an x^* to any level of accuracy.

Bisection method

The pesudo code of the bisection algorithm (sourced from Wikipedia) can be summarized as follows:

```
N ← 1
while N ≤ NMAX do // limit iterations to prevent infinite loop
    c ← (a + b)/2 // new midpoint
    if f(c) = 0 or (b – a)/2 < TOL then // solution found
        Output(c)
        Stop
    end if
    N ← N + 1 // increment step counter
    if sign(f(c)) = sign(f(a)) then a ← c else b ← c // new interval
end while
```

with

- f: the target function for which we want to find the root;
- $[a, b]$: the interval we start with bisecting;
- $NMAX$: the maximum number of iteration;
- TOL: the tolerance level of the half of the length of the returned interval in which the root lies.

Bisection Method for Implied Volatility

We will write our nonlinear equation for σ^* in the form $F(\sigma) = 0$, where

$$F(\sigma) := C(\sigma) - C^{\mathrm{mkt}}$$

and C^{mkt} is the market quote.

To apply the bisection method, we require an interval $[\sigma_a, \sigma_b]$ over which $F(\sigma)$ changes sign. We know theoretically this is possible, given that

$$\max\left(S_t - Ke^{-r(T-t)}, 0\right) \leq C(\sigma) < S_t$$

and that $C(\sigma)$ is monotonic.

Advantage: can be applied when there is no prior knowledge of the derivatives or initial guess of the objective function.

Newton's Method

The Newton's method (also called the Newton–Raphson method) can be derived in a number of ways. We will use the Taylor expansion approach.

Suppose we wish to compute a sequence $[x_0, x_1, x_2, \cdots]$ that converges to a solution x^* of $F(x^*) = 0$. We may expand $F(x_n + \delta)$ for small δ by

$$F(x_n + \delta) = F(x_n) + \delta F'(x_n) + O(\delta^2).$$

Ignoring the $O(\delta^2)$ term and setting $F(x_n + \delta) \approx F(x_n) + \delta F'(x_n) = 0$. It follows that if x_n is close to a solution x^* then

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$$

should be even closer to x^*.

Given a starting value x_0, the iteration

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$$

defines Newton's method.

Newton's Method

Theorem

Suppose that F has a continuous second derivative, and that there exists $x^* \in \mathbb{R}$ satisfying

$$F(x^*) = 0 \quad \text{and} \quad F'(x^*) \neq 0.$$

Then there exists a $\delta > 0$ such that for

$$|x_0 - x^*| < \delta,$$

the sequence defined by the Newton's method is well defined,

$$\lim_{n\to\infty} |x_n - x^*| = 0,$$

and there exists a constant C such that

$$|x_{n+1} - x^*| \leq C\,|x_n - x^*|^2.$$

Newton's Method

Intuitively, we can understand that the Newton's method has quadratic or second order convergence.

To see this, notice that

$$\begin{aligned} x_{n+1} - x^* &= x_n - \frac{F(x_n)}{F'(x_n)} - x^* \\ &= x_n - x^* - \frac{F(x_n) - F(x^*)}{F'(x_n)} \\ &= x_n - x^* - \frac{(x_n - x^*)F'(x_n) + O\big((x_n - x^*)^2\big)}{F'(x_n)} \\ &= O\big((x_n - x^*)^2\big) . \end{aligned}$$

where we used the fact that $F(x^*) = 0$ in the second line above and Taylor expansion in the third line above.

Advantage: faster convergence than bisection method
Disadvantage: depend on the initial guess

Newton's method to solve the implied volatility

The second order derivative of the call price with respect to the volatility is

$$\frac{\partial^2 C}{\partial \sigma^2} = \frac{S_t\sqrt{T-t}}{\sqrt{2\pi}} e^{-\frac{1}{2}d_1^2} \cdot \frac{d_1 d_2}{\sigma}, \qquad \sigma > 0.$$

Note that $d_1 d_2 = \frac{\ln(S_t/K)+(r+\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \cdot \frac{\ln(S_t/K)+(r-\frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$.

We have $\frac{\partial^2 C}{\partial \sigma^2} = 0$ when $d_1 d_2 = 0$, which holds if and only if

$$\sigma = \hat{\sigma} := \sqrt{\frac{2\,|\ln(S_t/K) + r(T-t)\,|}{T-t}}.$$

For $0 \leq \sigma < \hat{\sigma}$, $\frac{\partial^2 C}{\partial \sigma^2} > 0$, implying that $C(\sigma)$ is convex.
For $\sigma > \hat{\sigma}$, $\frac{\partial^2 C}{\partial \sigma^2} < 0$, implying that $C(\sigma)$ is concave.

If we start the Newton's method with the initial value $\sigma_0 = \hat{\sigma}$, a global convergence is guaranteed.

Newton's method to solve the implied volatility

Suppose the true implied volatility satisfies $\sigma^* > \sigma_0 = \hat{\sigma}$.

By Newton's method,

$$\sigma_{n+1} = \sigma_n - \frac{F(\sigma_n)}{F'(\sigma_n)} = \sigma_n - \frac{F(\sigma_n) - F(\sigma^*)}{F'(\sigma_n)}.$$

In particular,

$$\sigma_1 - \sigma^* = \sigma_0 - \sigma^* - \frac{F(\sigma_0) - F(\sigma^*)}{F'(\sigma_0)} = \sigma_0 - \sigma^* - \frac{(\sigma_0 - \sigma^*)F'(\xi_0)}{F'(\sigma_0)},$$

where the last equality follows from the Mean Value Theorem, for some ξ_0 satisfying

$$\sigma_0 < \xi_0 < \sigma^*.$$

Since $\sigma_0 = \hat{\sigma}$ maximizes $F'(\cdot)$, we have

$$\frac{\sigma_1 - \sigma^*}{\sigma_0 - \sigma^*} = 1 - \frac{F'(\xi_0)}{F'(\sigma_0)} \in (0, 1).$$

which implies $\sigma_0 < \sigma_1 < \sigma^*$.

Newton's method to solve the implied volatility

Next,

$$\sigma_2 - \sigma^* = \sigma_1 - \sigma^* - \frac{F(\sigma_1) - F(\sigma^*)}{F'(\sigma_1)} = \sigma_1 - \sigma^* - \frac{(\sigma_1 - \sigma^*)F'(\xi_1)}{F'(\sigma_1)},$$

for some ξ_1 satisfying $\sigma_1 < \xi_1 < \sigma^*$. It follows that

$$\frac{\sigma_2 - \sigma^*}{\sigma_1 - \sigma^*} = 1 - \frac{F'(\xi_1)}{F'(\sigma_1)} \in (0, 1).$$

Because $\partial C/\partial \sigma$ is positive and decreasing on $[\sigma_0, \infty)$ and $\sigma_0 < \sigma_1 < \xi_1 < \sigma^*$, it follows that $\sigma_0 < \sigma_1 < \sigma_2 < \sigma^*$.

Continuing this argument gives $\sigma_n < \sigma_{n+1} < \sigma^*$.
The same conclusion also holds if $\sigma^* < \sigma_0 = \hat{\sigma}$.

We conclude that with the choice $\sigma_0 = \hat{\sigma}$, the error decreases monotonically as the number of iterations increases. Hence the error tends to zero, and the convergence is quadratic.

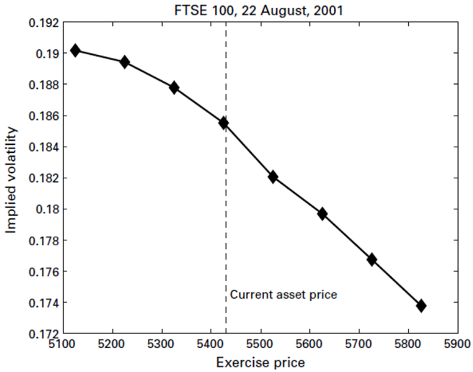
Implied volatility: Example

We look at the Implied volatility for call options traded on the London International Financial Futures and Options Exchange (LIFFE). The data is for the FTSE 100 index, which is an average of 100 equity shares quoted on the London Stock Exchange. The maturity date for these options was December 2001.
The initial asset price (on 22 August 2001) was 5420.3. We took values of $r = 0.05$ for the interest rate and $T = 1/3$ (years).

Exercise price	Option price
5125	475
5225	405
5325	340
5425	$280\frac{1}{2}$
5525	226
5625	$179\frac{1}{2}$
5725	139
5825	105

Implied volatility: Example

Implied volatility of FTSE 1000 As of Aug 22, 2001.



If the Black-Scholes formula is valid, the volatility should be the same for each exercise price. It means the the real market does not always perfectly follow the assumptions of the Black-Scholes model.

Implied volatility shapes (w.r.t. strike)

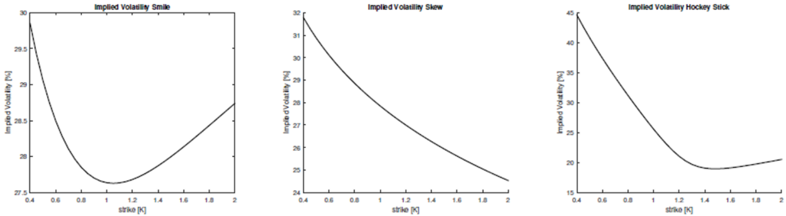


Figure: Typical implied volatility shapes: a smile, a skew and the so-called hockey stick. The hockey stick can be seen as a combination of the implied volatility smile and the skew.

Implied volatility term structure (w.r.t. maturity)

$$\frac{dS}{S} = \mu dt + \sigma(t) dW_t$$

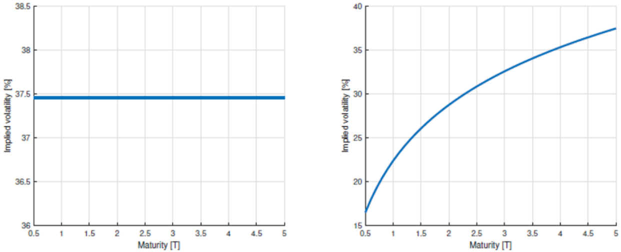


Figure: Comparison of the volatility term structure (for ATM volatilities) for Black-Scholes model with constant volatility σ_* (left-side picture) versus a model with time-dependent volatility $\sigma(t)$ (right-side picture).

Implied volatility surface

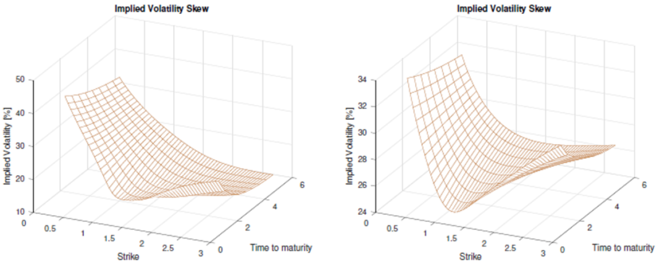


Figure: Implied volatility surfaces. A pronounced smile for the short maturity and a pronounced skew for longer maturities T (left side figure), and a pronounced smile over all maturities (right side).

Deficiencies of the Black-Scholes model

The Black-Scholes model is the foundation of option pricing theory.

However, some assumptions of the Black-Scholes model may not hold in reality:

- Trading (i.e. the self-financing trading strategy underlying delta hedging) is assumed to be a continuous time process while in reality it is a discrete process (modern trading strategies do allow intraday delta hedging);
- No transaction cost;
- The normality assumption of returns is in contrast with heavy tails in the data;
- the volatility parameter is constant; while volatility clustering is typical in financial returns.

Deficiencies of the Black-Scholes model

Other models to fit market quotes

- Term structure volatility: $\frac{dS}{S} = \mu dt + \sigma(t) dW_t$
- Local volatility models: $\frac{dS}{S} = \mu dt + \sigma(S, t) dW_t$
- Stochastic volatility models: $\frac{dS}{S} = \mu dt + \sigma(W^v, t) dW_t^S$
- Models with jumps: Will be discussed in later lectures.